6.10 FPGA Capacity: Max Gates vs Usable Gates

FPGA Capacity: Maximum Gates vs. Usable Gates

- Equivalent gate count:
  - the count of circuitry that can fit into a particular FPGA; useful information to designer
  - It is difficult to compute.
  - It is estimated in many different ways:
    - can be ascertained by considering typical ckts that can be implemented in a logic block.
      - E.g.: a 2-to-1 MUX is considered to be four gates
    - can be obtained by using ***benchmark circuits*** (✓)
      - **Programmable Electronics Performance Company (PREP)** benchmark suite

- It is challenging to estimate gate counts of FPGAs in which logic is implemented w/ ***LUTs***.
  - Vendors often compute their "system gates" count by considering a fraction of CLBs (say 20–30%) as RAM.
- Altera provides two types of gate counts for their APEX family: ***max gates*** and ***usable gates***.
- Many FPGA vendors provide their chip capacities w/ a count of the ***logic blocks*** (logic elements) rather than a gate count.